Benchmarking Machine Learning Interatomic Potentials and Classical Force Fields for Silicon and Germanium

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1 Introduction

Molecular Dynamics (MD) is a computational simulation methodology that utilizes the framework of Newtonian classical mechanics to model the physical movements of atoms and molecules over time, thereby providing atomistic-level detail of material behavior across diverse scientific and engineering domains [1, 2, 3]. The accuracy of any MD simulation fundamentally depends on the quality of the interatomic potential used to describe interactions between atoms, and the development of such potentials has remained a cornerstone challenge of computational materials science for over half a century [4]. Classical empirical potentials such as Tersoff [5, 6] and Stillinger–Weber (SW) [7] employ fixed analytical functional forms whose parameters are fitted to reproduce target properties from either quantum mechanical calculations, primarily Density Functional Theory (DFT), or experimental measurements. While computationally efficient, these potentials suffer from inherent limitations: their accuracy is bounded by the assumed functional form, they require compromises during parameter fitting, and they exhibit poor transferability to configurations outside their training domain.

To circumvent these limitations, Machine Learning Interatomic Potentials (MLIPs) have emerged as a transformative paradigm, replacing hand-crafted analytical functions with flexible, data-driven models that learn the complex, high-dimensional potential energy surface directly from *ab initio* data [8, 9, 10]. Among the most prominent MLIPs in current use are MACE (Multilayer Atomic Cluster Expansion) [11, 12], DeePMD-kit (Deep Potential Molecular Dynamics) [13, 14], and CHGNet (Crystal Hamiltonian Graph Neural Network) [15]. Each framework embodies distinct architectural innovations: MACE employs equivariant message passing combined with the Atomic Cluster Expansion formalism to capture higher-body-order interactions with exceptional efficiency; DeePMD-kit uses symmetry-preserving descriptors mapped through deep neural networks to per-atom energies; and CHGNet uniquely incorporates magnetic moments as proxies for electronic charge states, enabling charge-informed atomistic modeling.

In this work, we systematically benchmark four machine learning interatomic potentials, namely MACE MP 0 (pre-trained), CHGNet (pre-trained), DPA 3 SemiCond (pre-trained), and DeepMD (fine-tuned via Optuna hyperparameter optimization), alongside two classical interatomic potentials, namely Tersoff and Stillinger Weber, in their ability to predict the equilibrium lattice constant, cohesive energy, and internal force stability of silicon (Si) and germanium (Ge) in the diamond cubic crystal structure ($Fd\bar{3}m$). All predictions are compared against DFT (GGA-PBE) reference values obtained from the Materials Project database [17] (Si: mp-149, Ge: mp-32) and experimental measurements from crystallographic literature.

2 Methodology

2.1 Reference Data

Ground-truth reference values were sourced from two independent channels: (i) DFT calculations from the Materials Project [17] using the GGA-PBE exchange-correlation functional, and (ii)

experimental crystallographic data. The reference lattice constants are Si: $a_{\text{exp}} = 5.431\,\text{Å}$, $a_{\text{DFT}} = 5.469\,\text{Å}$ (mp-149); Ge: $a_{\text{exp}} = 5.658\,\text{Å}$, $a_{\text{DFT}} = 5.763\,\text{Å}$ (mp-32). Cohesive energy references are Si: $E_{\text{exp}} = -4.63\,\text{eV/atom}$, $E_{\text{DFT}} = -5.43\,\text{eV/atom}$; Ge: $E_{\text{exp}} = -3.85\,\text{eV/atom}$, $E_{\text{DFT}} = -4.62\,\text{eV/atom}$.

Table 1: Ground-truth reference values used for benchmarking.

Element	**Structure**	a_{exp} (Å)	a_{DFT} (Å)	E_{exp} (eV/atom)	E_{DFT} (eV/atom)
Si	Diamond cubic	5.431	5.469 (mp-149)	−4.63	−5.43
Ge	Diamond cubic	5.658	5.763 (mp-32)	−3.85	−4.62

2.2 Evaluated Potentials

2.2.1 Machine Learning Interatomic Potentials (MLIPs)

MACE-MP-0 (pre-trained): The MACE-MP-0 foundation model [12] was used directly without fine-tuning. It was pre-trained on the Materials Project Trajectory (MPtrj) dataset encompassing over 1.5 million inorganic structures and parameterized for 89 chemical elements. The model uses an equivariant message-passing architecture with ACE-based many-body descriptors.

CHGNet (pre-trained): The CHGNet v0.3.0 pre-trained model [15] was used in its foundation model configuration. It was pre-trained on the Materials Project Trajectory dataset with unique incorporation of magnetic moments to capture charge-state-dependent interactions.

DPA-3 SemiCond (pre-trained): The DPA-3 model from DeePMD-kit v3 with the "SemiCond" pre-trained head was selected specifically for semiconductor materials. It uses a DPA-2 descriptor architecture with multi-task learning. Per-atom energies use a different atomic reference convention ($\approx -107\,\text{eV/atom}$).

DeepMD `se_e2_a` (fine-tuned): This model was trained from scratch on MACE-teacher-generated data. The dataset comprised 200 training configurations per element (400 total). Optuna hyperparameter optimization was performed over 6 trials. The best trial (Trial 4) achieved $\text{RMSE}(F) = 0.0632\,\text{eV/Å}$, $r_{\text{cut}} = 6.0\,\text{Å}$, $\eta_0 = 3.34 \times 10^{-3}$, descriptor neurons [25, 50, 100], and 5000 training steps. Training was CPU-only.

2.2.2 Classical Interatomic Potentials

Tersoff: A bond-order potential capturing directional covalent bonds through coordination-dependent modulation of pairwise attraction. The LAMMPS parameterization for Si and Ge was used.

Stillinger–Weber (SW): A two-body plus three-body angular potential stabilizing tetrahedral coordination. It was used in place of ReaxFF because a suitable parameterization was unavailable for elemental Si/Ge in LAMMPS.

2.3 Benchmarking Protocol

2.3.1 Lattice Constant via EOS Fitting

Eleven diamond-cubic unit cells spanning ±6% around the experimental value were generated. Energy–volume data were fitted to the third-order Birch–Murnaghan equation of state. The equilibrium lattice constant was determined using $a_{\text{EOS}} = (4V_0)^{1/3}$.

2.3.2 Cohesive Energy via Full Relaxation

Primitive diamond-cubic cells were fully relaxed using ExpCellFilter + BFGS (ASE), with $f_{\max} = 0.01\,\text{eV/Å}$.

2.3.3 Force Stability via Bootstrap Random Displacements

Thirty configurations with random displacements in the range $[0.01, 0.20)$ Å were generated. The metric used was the mean and standard deviation of the maximum atomic force magnitude, $|F|_{\max}$.

2.3.4 Wide-Range Energy–Strain Curves

Twenty-one lattice constant values spanning 85% to 115% of the experimental value were evaluated.

Table 2: Best hyperparameters from Optuna HPO for DeepMD `se_e2_a`.

Parameter	**Value**
Cutoff radius (r_{cut})	6.0 Å
Smooth cutoff ($r_{\text{cut,smth}}$)	0.3 Å
Descriptor neurons	[25, 50, 100]
Fitting neurons	[240, 240, 240]
Initial learning rate (η_0)	3.34×10^{-3}
Decay steps	300
Total training steps	5000
Final RMSE(F)	0.0632 eV/Å

2.4 Computational Tools

- **ASE**: Structure generation, cell relaxation, EOS fitting
- **MACE** v0.3+: Pre-trained MACE-MP-0 via `mace.calculators.mace_mp`
- **CHGNet** v0.3.0: Pre-trained model via `chgnet.model.dynamics.CHGNetCalculator`
- **DeePMD-kit** v3: DPA-3 SemiCond; DeepMD `se_e2_a` training with Optuna
- **LAMMPS**: Tersoff and SW via `ase.calculators.lammpsrun.LAMMPS`
- **Optuna**: Bayesian HPO (6 trials)
- **Matplotlib** (STIX serif fonts): 300 DPI, PNG + PDF
- **Materials Project API**: DFT data for Si (mp-149) and Ge (mp-32)

2.5 Challenges and Resolutions

- **ReaxFF Parameterization Unavailability:** Reliable ReaxFF parameters for Si/Ge were not available in LAMMPS. Stillinger–Weber was used instead.
- **DPA-3 Multi-Task Head Extraction:** Custom scripts were required to parse the multi-task checkpoint and isolate the SemiCond head with proper buffer handling.
- **CUDA Errors:** Resolved by forcing CPU training.

- **DPA-3 Energy Reference:** Because of the different atomic reference ($\approx -107\,\mathrm{eV/atom}$), DPA-3 was excluded from energy parity analysis but retained for lattice and force benchmarks.
- **DeepMD Training Data:** MACE-teacher knowledge distillation (400 configurations) introduces an accuracy ceiling.
- **MEAM Failure:** Library format incompatibilities led to replacement by SW.

3 Results

3.1 Lattice Constant Predictions

CHGNet achieves the highest accuracy: 0.09% (Si) and 0.08% (Ge). MACE-MP-0 yields 0.23% (Si) and 0.12% (Ge). DPA-3 gives 0.27% for both systems. DeepMD gives 2.68% (Si) and 1.60% (Ge). Tersoff produces 0.69% (Si) and 1.84% (Ge), while SW gives 0.70% (Si) and 1.90% (Ge). Overall parity metrics are $R^2 = 0.7945$, MAE $= 0.0492\,\text{Å}$, and RMSE $= 0.0688\,\text{Å}$ over 12 data points. Excluding DeepMD, $R^2 > 0.95$.

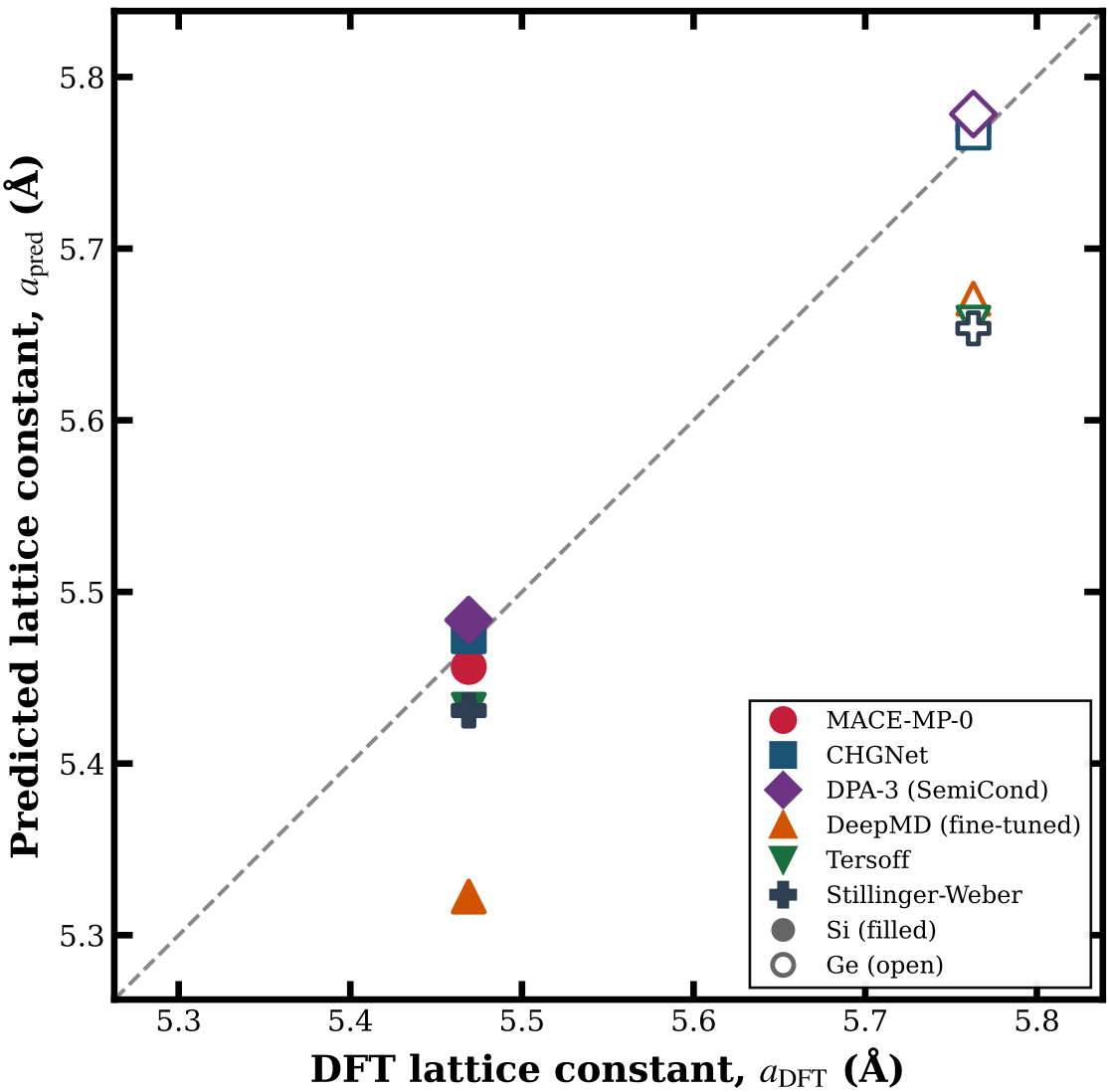

Figure 1: Parity plot of predicted versus DFT equilibrium lattice constants for six potentials on Si and Ge. Filled markers denote Si; open markers denote Ge. The dashed line indicates perfect agreement.

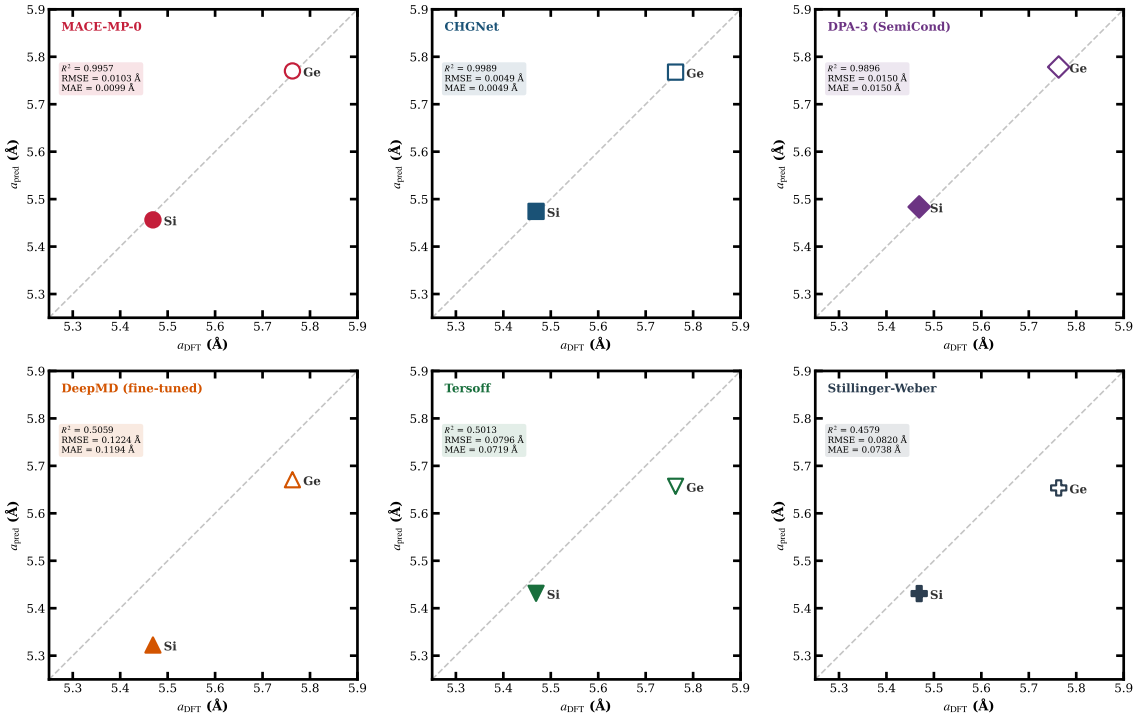

Figure 2: Individual lattice-constant parity plots (2×3 grid) with MAE, RMSE and R^2. CHGNet shows the best MAE at 0.0049 Å.

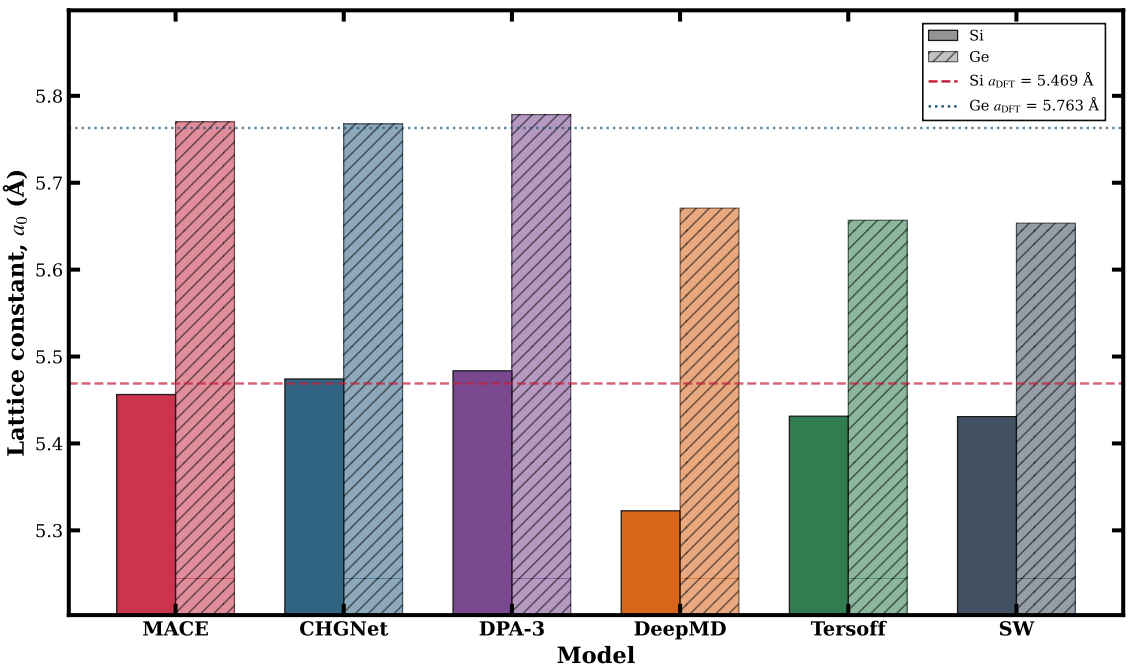

Figure 3: Grouped bar chart of lattice constants with DFT reference lines.

3.2 Cohesive Energy Predictions

DPA-3 is excluded because it uses a different atomic reference. Across the remaining models, $R^2 = -0.0878$, MAE = 0.3996 eV, and RMSE = 0.5540 eV. MACE-MP-0 predicts Si at −5.342 eV/atom ($\Delta = 1.6\%$) and Ge at −4.526 eV/atom ($\Delta = 2.0\%$). Classical potentials underpredict cohesive energies by 14–20%.

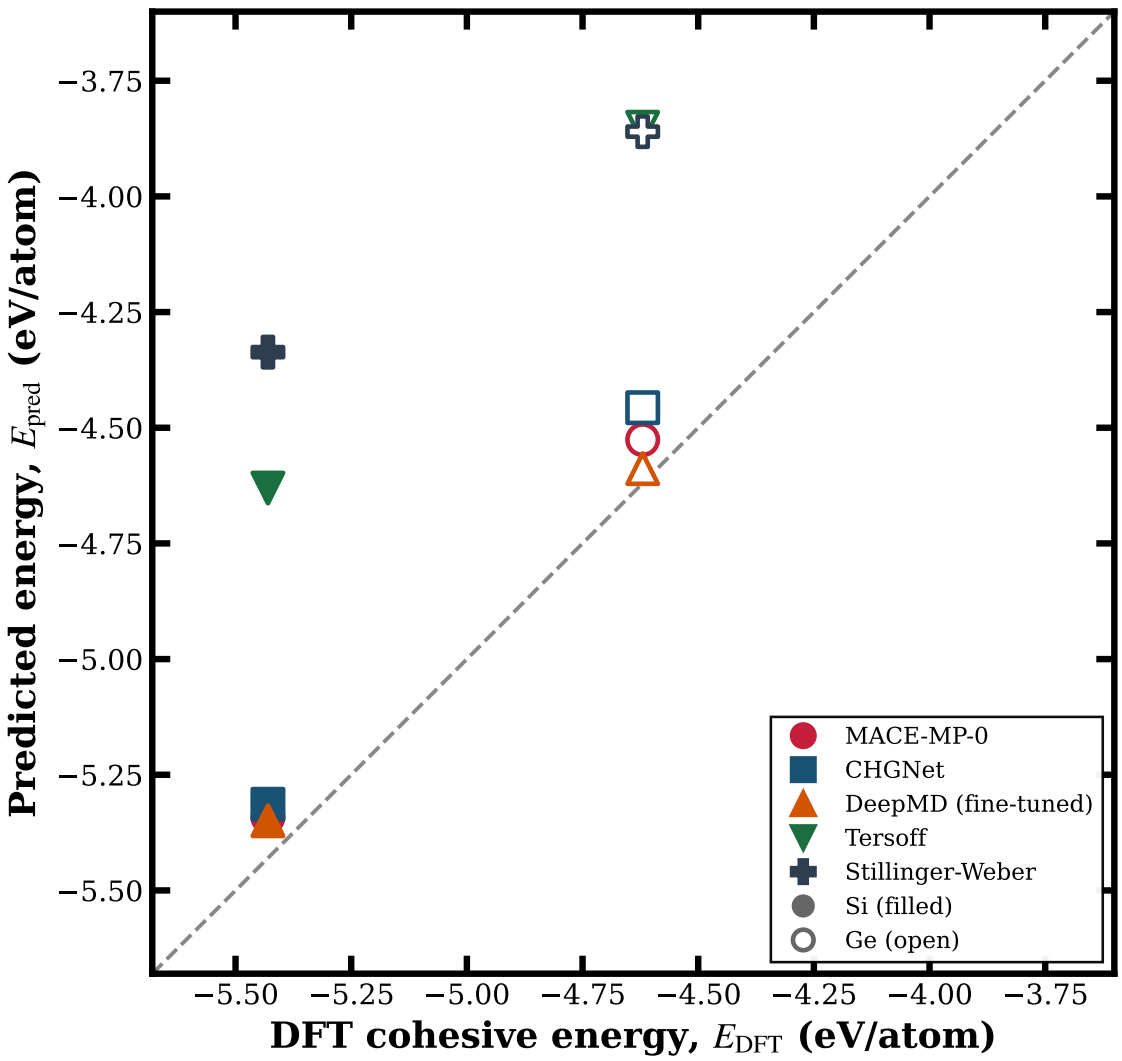

Figure 4: Cohesive energy parity plot for five models, with DPA-3 excluded.

3.3 Equation of State

Pre-trained MLIPs exhibit smooth, symmetric parabolic wells near the DFT lattice constant. DeepMD shows an asymmetric well for Si and an anomalous 8 eV/atom spike under compression. Tersoff and SW exhibit broader, shallower wells.

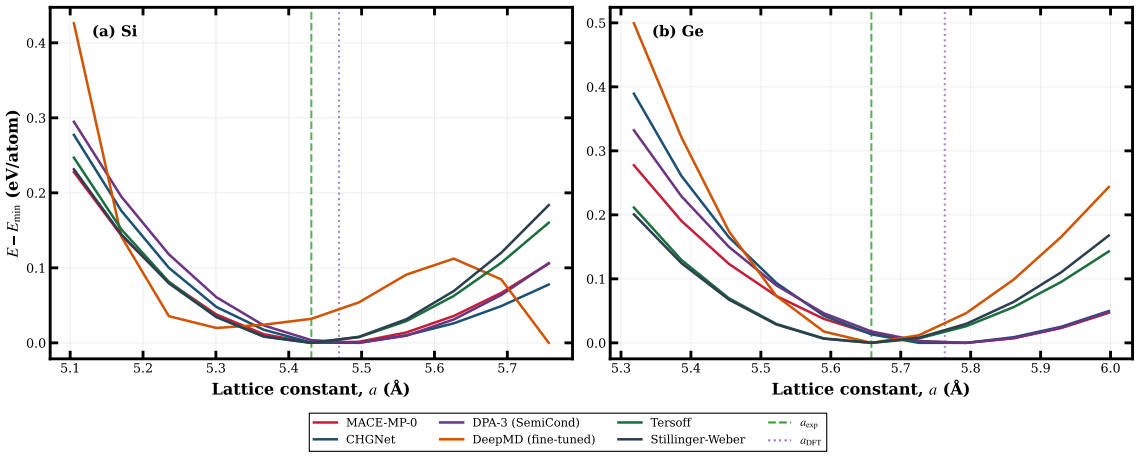

Figure 5: Equation-of-state curves for Si (a) and Ge (b), plotting $E - E_{\min}$ versus lattice constant.

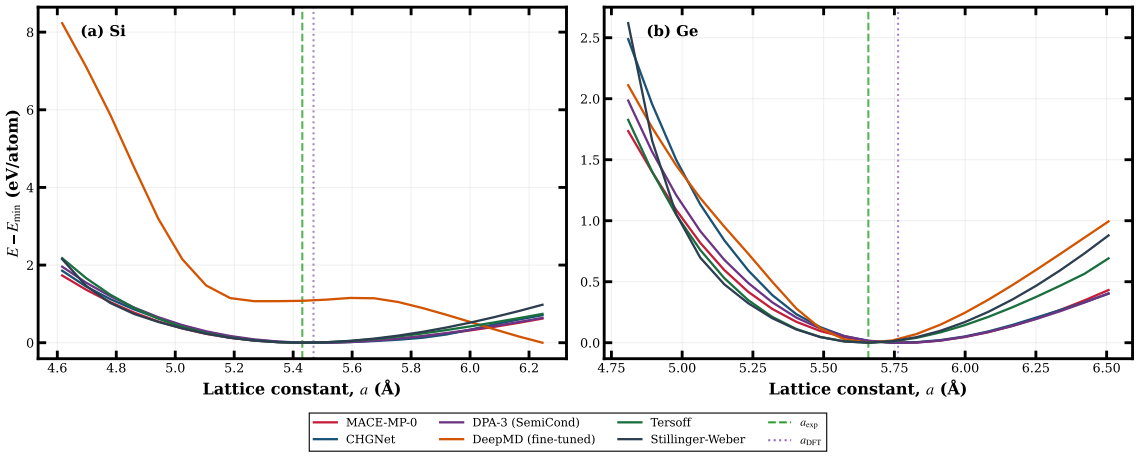

Figure 6: Wide-range energy–strain curves over 85–115% of $a_{\exp}$.

3.4 Force Stability

MLIPs produce $\langle |F|_{\max} \rangle \leq 2.3\,\mathrm{eV/Å}$. DeepMD is the smoothest, with $1.69 \pm 1.93\,\mathrm{eV/Å}$ for Si and $1.12 \pm 1.22\,\mathrm{eV/Å}$ for Ge. Classical potentials are roughly two to three times higher. Tersoff-Si exhibits $\sigma = 5.24\,\mathrm{eV/Å}$, suggesting cutoff-induced spikes.

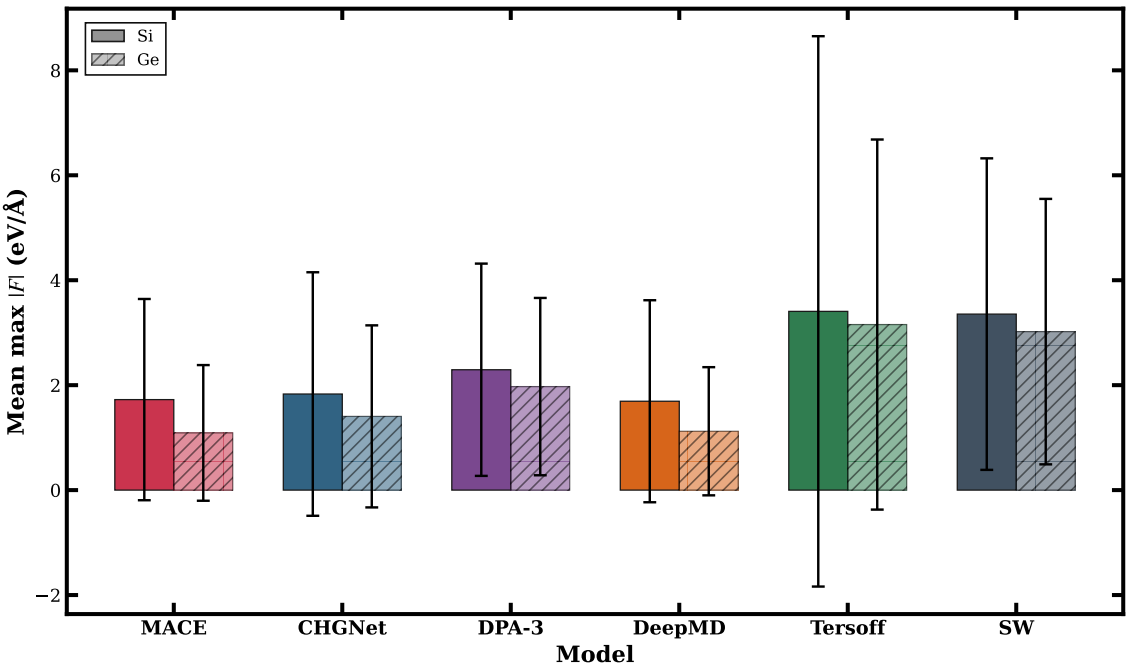

Figure 7: Force stability measured as the mean maximum $|F|$ over 30 random displacements.

3.5 Combined Dashboard and Optuna

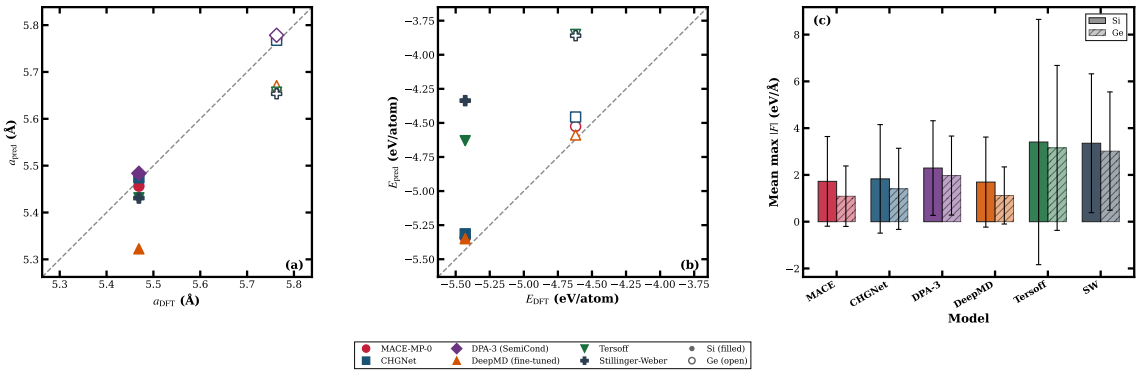

Figure 8: Combined 1×3 dashboard showing (a) lattice parity, (b) energy parity, and (c) force stability.

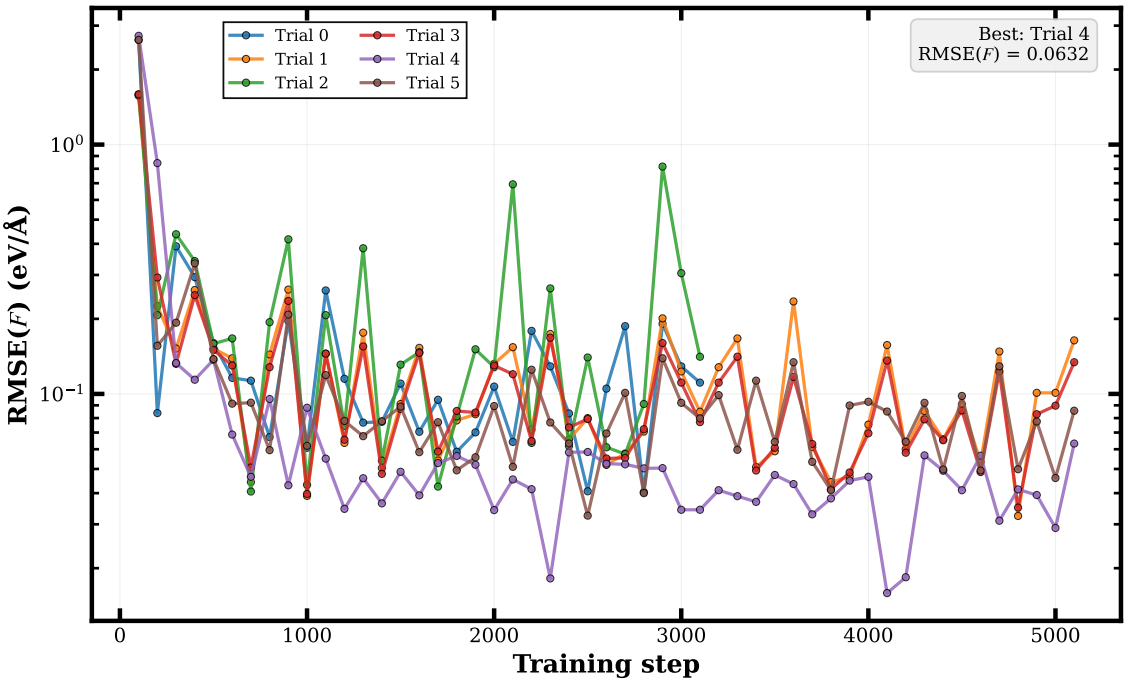

Figure 9: DeepMD Optuna HPO convergence over 6 trials. The best result was Trial 4 with $\mathrm{RMSE}(F) = 0.0632\,\mathrm{eV/Å}$.

4 Comprehensive Benchmark Summary

Table 3: Summary of benchmark results for six interatomic potentials on Si and Ge.

Model	**El**	a_{EOS} (Å)	Δa (%)	E_{rlx} (eV/atom)	$\langle \lvert F \rvert_{\max} \rangle$ (eV/Å)
MACE-MP-0	Si	5.4563	0.23	−5.342	1.72 ± 1.92
	Ge	5.7701	0.12	−4.526	1.09 ± 1.29
CHGNet	Si	5.4742	0.09	−5.314	1.83 ± 2.32
	Ge	5.7677	0.08	−4.457	1.41 ± 1.73
DPA-3 (SemiCond)	Si	5.4836	0.27	−107.21*	2.29 ± 2.02
	Ge	5.7784	0.27	−107.10*	1.97 ± 1.69
DeepMD (fine-tuned)	Si	5.3225	2.68	−5.349	1.69 ± 1.93
	Ge	5.6708	1.60	−4.588	1.12 ± 1.22
Tersoff	Si	5.4313	0.69	−4.630	3.41 ± 5.24
	Ge	5.6569	1.84	−3.851	3.16 ± 3.53
Stillinger–Weber	Si	5.4309	0.70	−4.337	3.35 ± 2.97
	Ge	5.6535	1.90	−3.860	3.02 ± 2.53

5 Discussion: Model Rankings

Table 4: Ranking of interatomic potentials by average lattice-constant accuracy and force stability.

Rank	Model	Avg $\lvert\Delta a\rvert$ (%)	Avg $\langle \lvert F\rvert_{\max} \rangle$	Type
1	CHGNet	0.088	1.62	MLIP
2	MACE-MP-0	0.178	1.41	MLIP
3	DPA-3 (SemiCond)	0.267	2.13	MLIP
4	Tersoff	1.265	3.28	Classical
5	Stillinger–Weber	1.299	3.19	Classical
6	DeepMD (fine-tuned)	2.140	1.41	MLIP

Best Classical Potential

Tersoff is marginally preferred: it provides comparable lattice accuracy to SW but more accurate cohesive energies (Si: -4.630 versus $-4.337\,\mathrm{eV/atom}$; 14.7% versus 20.1% error). SW has lower force variance ($\sigma = 2.97$ versus $5.24\,\mathrm{eV/Å}$ for Si).

Verdict: Tersoff for accuracy; SW for stable MD dynamics.

Best MLIP

Overall Verdict:

- **Best overall: CHGNet** — 0.088% lattice error, competitive energy, and acceptable forces
- **Best energy: MACE-MP-0** — 1.6–2.0% energy error, 0.178% lattice error, and best force stability
- **Best for MD: MACE-MP-0** — smoothest PES with $\langle \lvert F\rvert_{\max} \rangle = 1.41\,\mathrm{eV/Å}$
- **DPA-3** — solid all-around performance with specialized semiconductor training
- **DeepMD** — demonstrates knowledge-distillation viability but poor structural accuracy

6 Conclusion

Pre-trained MLIP foundation models (CHGNet with sub-0.1% lattice error; MACE-MP-0 with approximately 2% energy error) substantially outperform classical potentials (Tersoff/SW with 0.7–1.9% lattice error and 14–20% energy error) for Si and Ge. Neural-network-based potentials produce force landscapes that are roughly two to three times smoother. The DeepMD MACE-teacher distillation approach is viable but inherits accuracy ceilings. These results demonstrate that foundation-model MLIPs trained on more than 1.5 million DFT structures can achieve near-DFT accuracy without system-specific training.

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